

Tail-Aware Post-Training Quantization for 3D Geometry Models

Anonymous Author(s)

Submission Id: 3384

Abstract

The burgeoning complexity and scale of 3D geometry models pose significant challenges for deployment on resource-constrained platforms. While Post-Training Quantization (PTQ) enables efficient inference without retraining, existing PTQ methods for 2D Vision Transformers do not readily extend to 3D models, due to more extreme heavy-tailed feature distributions from multi-view modeling and the prohibitive cost of traditional calibration interval search algorithms. To address these challenges, we propose **TAPTQ**, a Tail-Aware Post-Training Quantization pipeline specifically engineered for 3D geometric learning. Our contribution is threefold: (1) To overcome the data-scale bottleneck in 3D datasets, we develop a progressive coarse-to-fine calibration construction strategy that constructs a highly compact subset to achieve both statistical purity and geometric representativeness. (2) We reformulate the quantization interval search as an optimization problem and introduce a ternary-search-based solver, reducing the computational complexity from $O(N)$ to $O(\log N)$ for accelerated deployment. (3) To mitigate quantization error accumulation, we propose TRE-Guided Module-wise Compensation, which utilizes a Tail Relative Error (TRE) metric to adaptively identify and rectify distortions in modules sensitive to long-tailed activation outliers. Extensive experiments on the VGGT and Pi3 benchmarks demonstrate that TAPTQ consistently outperforms state-of-the-art PTQ methods in accuracy while significantly reducing calibration time. The code will be released soon.

CCS Concepts

• Computing methodologies → 3D imaging.

Keywords

3D Geometry Models, Post-Training Quantization, Tail-Aware Quantization, Ternary Search, Tail Relative Error

1 Introduction

The emergence of large-scale 3D foundation models [10, 15, 24, 25] has revolutionized the field of computer vision by enabling end-to-end geometric learning directly from massive datasets. This paradigm shift addresses the fundamental challenge of inferring 3D structures from 2D images, which is a cornerstone task that has traditionally relied on Structure-from-Motion (SfM) pipelines [3, 20]. By employing unified, feed-forward architectures, these modern models jointly solve diverse tasks like multi-view reconstruction and camera estimation with unprecedented accuracy. These advances hold immense potential for real-world applications in augmented reality (AR), autonomous driving, and robotics. However, the superior performance of these models is inextricably tied to their burgeoning scale, which results in massive computational and memory footprints. This presents significant hurdles for practical deployment on resource-constrained or latency-sensitive edge platforms.

Post-training Quantization (PTQ) [5, 27, 32] has emerged as a pivotal technique for model compression, facilitating the conversion of high-precision weights and activations into low-bit integer arithmetic without exhaustive retraining. While PTQ has achieved remarkable success in Large Language Models (LLMs) and 2D Vision Transformers (ViTs), a systematic investigation into its efficacy for 3D geometry foundation models remains largely underexplored. Our experimental results show that directly transplanting established PTQ recipes from the 2D vision to 3D geometry transformers leads to severe performance degradation and prohibitive calibration overhead. Such failures suggest that the unique properties and multi-view dependencies of 3D geometric data require a specialized quantization paradigm that goes beyond existing general-purpose frameworks.

We attribute these observed failures to the inherent differences of both 3D geometric data and model, which manifest as three primary obstacles: **First, statistical heterogeneity in multi-view calibration.** Unlike 2D image tasks, a single calibration instance in 3D geometry models typically comprises multiple correlated yet non-identical views. The resulting activation distributions are highly sensitive to viewpoint configurations and noisy geometric structures, leading to substantial inter-sample statistical variance. Consequently, a sparse calibration set may fail to encapsulate the underlying data manifold, causing range estimation to be biased by unstable outliers. While recent work [4] attempts to mitigate this through model-dependent filtering, such approaches often necessitate auxiliary forward passes and lack cross-architecture portability. **Second, the computational bottleneck of range optimization.** High-fidelity PTQ relies on optimizing clipping ranges or scaling factors via iterative evaluations. In 3D geometry settings, this cost is severely exacerbated by the multi-view inference paradigm and massive token sequences (e.g., dense point maps), rendering standard $O(N)$ grid searches or reconstruction-based optimizations (e.g., BRECQ [12]) computationally prohibitive. This necessitates a more efficient, ideally logarithmic-complexity search paradigm to accelerate calibration without sacrificing quantization fidelity. **Third, intensified error sensitivity in multi-task geometric inference.** Unlike previous architecture (e.g., 2D classification via ViTs), which exhibits higher tolerance to quantization noise, 3D geometry foundation models demand significantly more stringent numerical precision. In complex tasks such as camera pose estimation and dense reconstruction, even minor quantization errors can amplify across transformer layers, leading to significant geometric drift and misalignment. This necessitates a principled and lightweight metric to selectively identify and rectify “bottleneck” modules to ensure geometric fidelity with minimal compensation overhead.

To address these challenges, we propose **TAPTQ** (Tail-Aware Post-Training Quantization), a high-efficiency pipeline specifically tailored for 3D geometry models. TAPTQ is built upon three core innovations: (i) **Progressive calibration set construction:** To

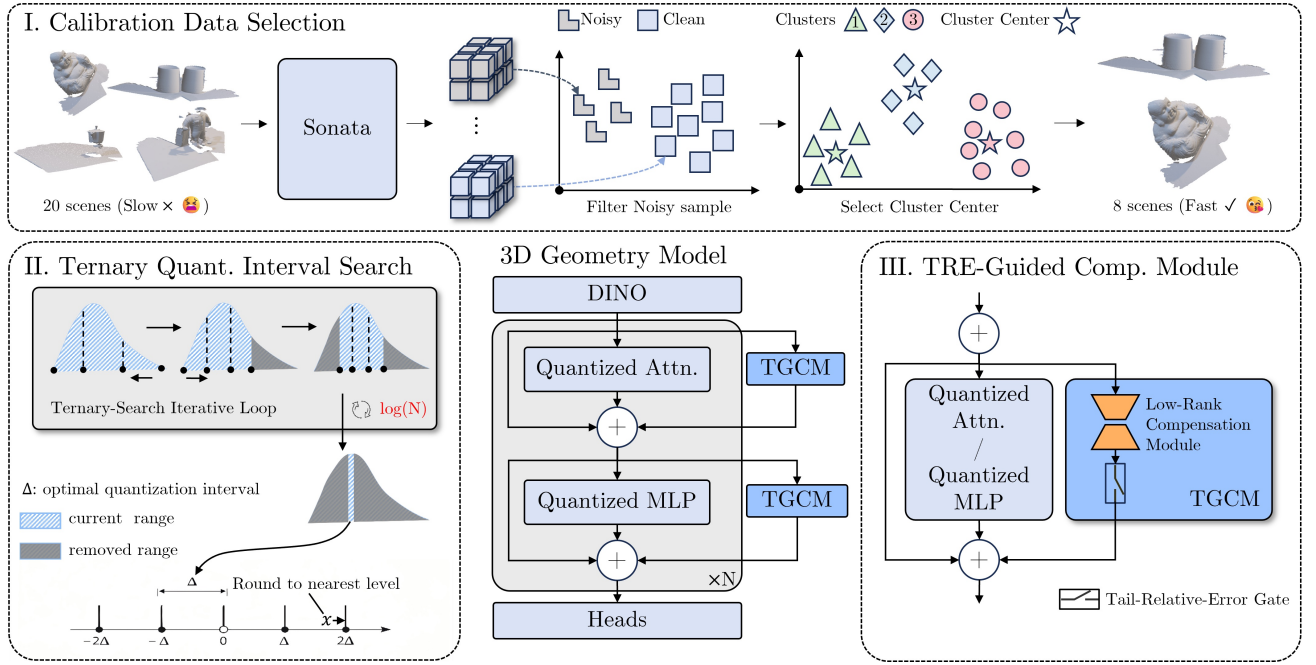


Figure 1: Overview of TAPTQ. (I) Representative Calibration Data via Two-Stage Cluster. (II) Quantization Interval Optimization via Ternary Search. (III) Tail-Relative-Error (TRE)-Guided Compensation Module.

ensure robustness under constrained data budgets, we develop a progressive coarse-to-fine calibration construction strategy to construct a compact yet statistically representative calibration set. This strategy suppresses the influence of unstable samples and noisy geometric structures, providing a solid foundation for consistent range estimation. **(ii) Ternary-search-based interval optimization:** We reformulate quantization interval selection as an optimization problem. By exploiting the near-unimodal property of the calibration objective, we introduce a ternary-search-based solver that reduces the search complexity from $O(N)$ to $O(\log N)$, significantly accelerating the deployment process without compromising quantization fidelity. **(iii) TRE-Guided Module-wise Compensation:** To mitigate error accumulation across transformer layers, we propose a selective compensation mechanism. We introduce the Tail Relative Error (TRE) metric to quantify distortions in high-magnitude activation regions. TRE acts as a gating criterion to adaptively identify and rectify only the most affected “bottleneck” modules, preserving geometric consistency with minimal computational overhead.

Our contributions are summarized as follows:

- We analyze why existing PTQ pipelines for LLMs and 2D vision transformers do not directly translate to 3D geometry transformers, emphasizing calibration fragility, interval-search cost, and the need for selective error correction.
- We adopt a progressive coarse-to-fine calibration construction strategy that constructs a compact yet robust calibration set under limited calibration budgets.

- We introduce a ternary-search-based interval calibration method that reduces the interval-search complexity from $O(N)$ to $O(\log N)$.
- We propose TRE-Guided Module-wise Compensation, using Tail Relative Error to selectively activate compensation and mitigate cross-layer error accumulation with minimal overhead.

2 Related Work

2.1 Feed-forward 3D Geometry Models

Traditional 3D reconstruction typically follows a two-stage paradigm: Structure-from-Motion (SfM) for pose estimation followed by Multi-View Stereo (MVS) for dense geometry [20, 31]. While neural radiance fields (NeRF) [2, 17] and 3D Gaussian Splatting [9, 29, 30] have revolutionized high-fidelity rendering, they fundamentally rely on *per-scene optimization*, which is computationally prohibitive for large-scale, real-time applications. To circumvent this, recent research has shifted toward *feed-forward geometry foundation models*. DUST3R [25] and MAST3R [10] pioneered direct geometry inference from unconstrained image collections by predicting dense pointmaps. VGGT [24] further unified multiple geometric tasks within a monolithic transformer architecture. Subsequent works have explored architectural efficiency through permutation-equivariant designs [26], unified depth-ray prediction [15], and attention sparsification [21, 23]. Despite these algorithmic advances, the massive parameter count and memory footprint of these transformers remain a bottleneck for edge deployment, necessitating effective model compression.

2.2 Post-Training Quantization (PTQ)

PTQ enables model compression using limited calibration data without the need for high-cost quantization-aware training (QAT) [7]. Early methods focused on MSE-driven range clipping [1] or weight equalization [19]. To achieve lower bit-widths, reconstruction-based approaches like AdaRound [18], BRECQ [12], and QDrop [27] optimize rounding or local block-wise behavior to minimize output distortion. The shift toward Transformer architectures introduced new challenges, specifically regarding non-linearities (Softmax/GELU) and heavy-tailed activation distributions. In the vision domain, methods such as PTQ4ViT [32] and RepQ-ViT [14] address these through specialized scaling and re-parameterization. Similarly, for Large Language Models (LLMs), frameworks like GPTQ [5], SmoothQuant [28], and AWQ [16] leverage one-shot weight updates or activation migration to preserve model capacity. However, recent analysis [8] suggests that the efficacy of these methods is highly sensitive to the choice of calibration data. Quantizing large-scale 3D geometry transformers is an emergent and under-explored area. While QuantVGTT [4] provided a preliminary study on the sensitivity of 3D activations, it does not fully address the calibration inefficiency and the complex error accumulation inherent in multi-view geometry tasks. Unlike general-purpose PTQ for 2D ViTs or LLMs, our work explicitly targets the unique failure modes of 3D models—specifically the tail-dominated activation ranges and view-dependent calibration variability. By introducing a ternary-search-based optimization and a tail-relative error metric, we bridge the gap between high-performance geometry foundation models and efficient hardware deployment.

3 Method

3.1 Preliminary

Post-training quantization (PTQ) compresses a pretrained model into low-bit arithmetic without retraining, by quantizing weights and activations using a small unlabeled calibration set. In this work, we adopt twin uniform quantization [32], which maps a real-valued tensor x to its quantized version x_q via

$$x_q = \begin{cases} \text{clip}\left(\left\lfloor \frac{x}{\Delta_{R_1}} \right\rfloor, q_{\min}, q_{\max}\right) \cdot \Delta_{R_1}, & x \in \mathcal{R}_1, \\ \text{clip}\left(\left\lfloor \frac{x}{\Delta_{R_2}} \right\rfloor, q_{\min}, q_{\max}\right) \cdot \Delta_{R_2}, & x \in \mathcal{R}_2. \end{cases} \quad (1)$$

where $\Delta > 0$ denotes the quantization interval (i.e., scaling factor) and $[q_{\min}, q_{\max}]$ are quantization ranges that are determined by the given bit-width setting.

The central goal of PTQ is to determine Δ , because a small interval leads to saturation while a large interval increases rounding error [22]. PTQ therefore determines Δ by comparing the outputs of the quantized model with those of the full-precision model on calibration data. Given the calibration inputs $\mathbf{x}$ and the model $\mathcal{F}(\cdot)$, let $\mathbf{y}_{\text{FP}} = \mathcal{F}(\mathbf{x})$ be the full-precision output and $\mathbf{y}_{\Delta} = \mathcal{F}(\mathbf{x}; \Delta)$ the corresponding quantized output using the quantization intervals Δ . Quantization interval optimization can be formulated as

$$\alpha^* = \arg \max_{\alpha \in \mathcal{A}} \text{Sim}(\mathbf{y}_{\text{FP}}, \mathbf{y}_{\alpha}), \quad (2)$$

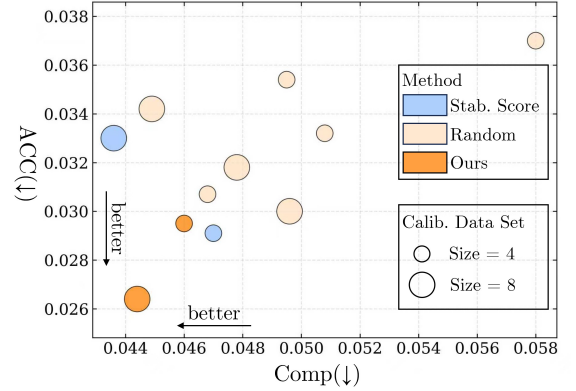


Figure 2: Acc. and Comp. performance under different calibration data set. Each bubble corresponds to one method configuration on 7Scenes (W4A8), where bubble size indicates the number of calibration samples (4 / 8 / 20 scans). Stab. score denotes a stability-based calibration sample selection strategy.

where $\mathcal{A} = \{\alpha_i\}_{i=1}^N$ is a discrete set of candidate intervals and $\text{Sim}(\cdot)$ denotes a similarity metric, defined as

$$\text{Sim}(\mathbf{y}_{\text{FP}}, \mathbf{y}_{\alpha}) = -\|\mathbf{g} \odot (\mathbf{y}_{\text{FP}} - \mathbf{y}_{\alpha})\|_2^2, \quad (3)$$

where $\mathbf{g}$ is the gradient associated with the corresponding tensor, weighting quantization errors by local sensitivity.

Previous works [11, 32] evaluate all candidates in $\mathcal{A}$ and thus incur $O(N)$ forward passes, which becomes a major calibration bottleneck for large 3D geometry models. This motivates efficient interval search strategies for PTQ, which we address in the following section.

3.2 Progressive Calibration Set Construction

Post-training quantization relies heavily on the quality of the calibration set to estimate accurate quantization ranges. Conventional PTQ frameworks typically resort to random sampling [5, 27]; however, we observe that for 3D geometry foundation models, such a stochastic approach leads to significant performance fluctuations. As demonstrated in Fig. 2, repeated random samplings result in a high variance of quantization accuracy, indicating that *unstable or noisy 3D samples can disproportionately distort the estimated quantization ranges*. To address this, we propose a progressive coarse-to-fine calibration construction strategy to achieve both *statistical purity* and *geometric representativeness*.

Stage 1: Coarse-grained Outlier Suppression. The initial stage aims to purify the calibration pool by filtering out noisy instances. Given an unlabeled dataset $\mathcal{D} = \{x_i\}_{i=1}^M$, we first map each 3D sample x_i to a latent feature space $\mathcal{Z}$ using a self-supervised point cloud encoder $f_{\theta}(\cdot)$, such that $z_i = f_{\theta}(x_i) \in \mathbb{R}^d$. We perform K -means clustering on $\mathcal{Z}$ with $K = 2$ to partition the dataset into two clusters, C_1 and C_2 .

Guided by the Pareto Principle in large-scale datasets, we hypothesize that high-quality samples constitute the statistical majority,

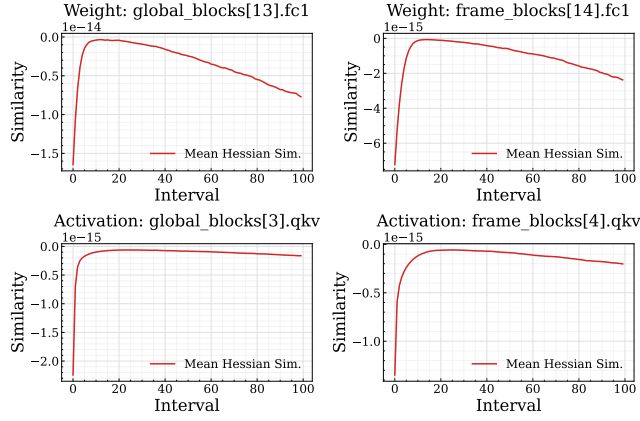


Figure 3: Unimodal behavior of similarity with respect to quantization interval. The similarity between quantized and full-precision outputs is evaluated over a discrete set of interval candidates for both activations and weights. In both cases, the similarity exhibits an approximately unimodal trend, enabling efficient interval search via ternary search.

whereas noisy or degenerate samples reside in a sparse minority cluster. We select the purified candidate set $\mathcal{D}_{pure}$ based on a majority-voting mechanism:

$$\mathcal{D}_{pure} = \arg \max_{C \in \{C_1, C_2\}} |C|, \quad (4)$$

where $|C|$ denotes the cardinality of the cluster. This procedure effectively suppresses the influence of unstable outliers, providing a robust distribution for subsequent selection.

Stage 2: Fine-grained Geometric Representation. After denoising, we seek to construct a final calibration set $\mathcal{C}$ ($|\mathcal{C}| = N$) that captures both diverse geometric configurations and local statistical variations. We perform fine-grained clustering on the features of $\mathcal{D}_{pure}$ with $K = N/2$ clusters, denoted as $\{S_k\}_{k=1}^{N/2}$. For each cluster S_k with centroid μ_k , we select two representative samples closest to the cluster centroid, defined as:

$$z_{k,1} = \arg \min_{z \in S_k} \|z - \mu_k\|_2. \quad (5)$$

$$z_{k,2} = \arg \min_{z \in S_k \setminus \{z_{k,1}\}} \|z - \mu_k\|_2. \quad (6)$$

The final calibration set is defined as $\mathcal{C} = \bigcup_{k=1}^{N/2} \{z_{k,1}, z_{k,2}\}$. This strategy ensures that $\mathcal{C}$ not only spans the broad geometric manifold but also encapsulates local variations, thereby enhancing the stability of the estimated quantization parameters.

3.3 Quantization Interval Optimization via Ternary Search

In conventional PTQ frameworks, interval selection is typically treated as an exhaustive grid search over a discrete candidate set $\mathcal{A} = \{\alpha_i\}_{i=1}^N$. This incurs prohibitive $O(N)$ computational overhead, as each candidate requires a complete quantized forward pass. For large-scale 3D geometry models, this linear complexity becomes a critical calibration bottleneck, with execution times often exceeding 522.6 GPU-minutes per model. To alleviate this, we propose an

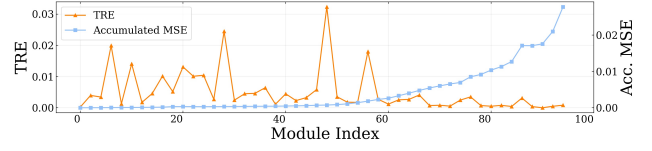


Figure 4: Accumulated MSE and Tail Relative Error across W4A8 quantized Transformer modules. The accumulated MSE is measured at the output of each quantized module along the network depth. Results are reported for a quantized VGGT model with layer-wise calibration applied.

accelerated interval optimization strategy based on the *unimodality* of the quantization objective.

Unimodality of the Similarity Landscape. We investigate the behavior of the similarity function $\text{Sim}(\mathbf{y}_{FP}, \mathbf{y}_\alpha)$ across the candidate space $\mathcal{A}$. As illustrated in Fig. 3, the similarity score exhibits a consistent *near-unimodal* trend for both weights and activations. This phenomenon is rooted in the fundamental trade-off of quantization: a narrow interval leads to excessive *clipping distortion*, while an overly wide interval increases *rounding noise* by coarsening the quantization step size. The competition between these two error sources naturally induces a unimodal landscape, where a unique global optimum exists.

Ternary Search Solver. Leveraging the unimodal property, we reformulate interval selection as a constrained optimization problem. Instead of a linear scan, we employ a ternary search algorithm to progressively narrow the search boundaries $[L, R]$ within the discrete index range $[1, N]$. At each iteration, we evaluate two interior points m_1 and m_2 :

$$m_1 = L + \frac{R - L}{3}, \quad m_2 = R - \frac{R - L}{3}. \quad (7)$$

By comparing the similarity scores Sim_{m_1} and Sim_{m_2} obtained from the quantized forward passes, the search space is updated as follows:

$$[L, R] \leftarrow \begin{cases} [m_1, R] & \text{if } \text{Sim}_{m_1} < \text{Sim}_{m_2}, \\ [L, m_2] & \text{otherwise.} \end{cases} \quad (8)$$

The search terminates when $R - L \leq \epsilon$, where ϵ is a small convergence threshold (e.g., $\epsilon = 0.0001$). This strategy is applicable to both weight and activation calibration, ensuring broad portability across different layers.

Complexity Analysis. Compared with conventional linear search, which requires $O(N)$ evaluations over the candidate set, our ternary-search-based solver reduces the search complexity to $O(\log N)$. In practice, this logarithmic reduction substantially alleviates the calibration overhead and makes interval optimization considerably more efficient.

Practical Optimization Details. In practice, to further improve robustness, we adopt an alternating optimization scheme between weight and activation intervals for a small number of rounds (e.g., 3 rounds). Moreover, when the search interval shrinks below a small threshold (e.g., 8 candidates), we perform exhaustive evaluation within the remaining candidate set to select the best interval. This coarse-to-fine strategy stabilizes the optimization and mitigates potential suboptimal selections in discrete search spaces.

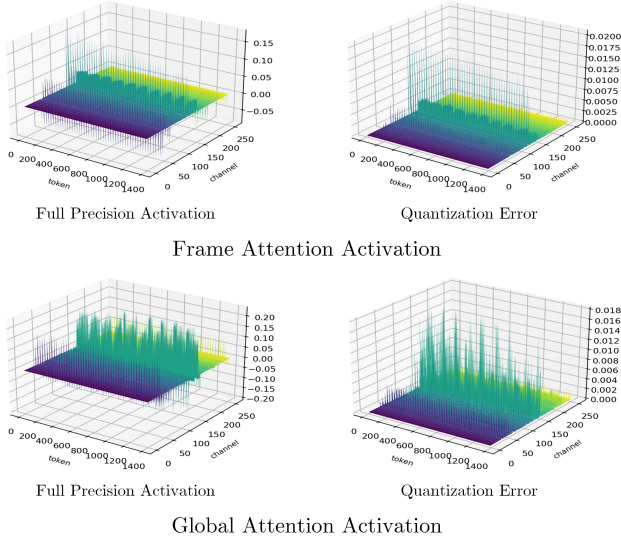


Figure 5: Visualization of activation values and corresponding quantization errors. Each horizontal pair shows the full-precision activations (left) and the corresponding quantization errors (right) for representative attention modules. Both distributions exhibit heavy-tailed characteristics, with large-magnitude activations contributing disproportionately to the observed quantization error.

3.4 TRE-Guided Module-wise Compensation

Despite optimal interval selection, PTQ for deep 3D geometry transformers still suffers from *cascading quantization noise*. Since PTQ follows a sequential calibration paradigm without end-to-end optimization, quantization errors introduced in preceding layers inevitably accumulate and amplify as they propagate through the network hierarchy. As illustrated in Fig. 4, the cumulative error at each transformer block’s output increases monotonically with depth. For dense geometric tasks, these amplified distortions lead to significant structural drift in the inferred 3D space. To mitigate this, we propose **TRE-Guided Module-wise Compensation**, a selective, low-rank correction mechanism.

Low-rank Residual Compensation. We apply compensation to the two core residual submodules in each transformer block: the Attention layer and the MLP layer. To rectify the residual error between the full-precision output y and its quantized counterpart y_q , we introduce a parallel lightweight branch that is inspired by QwT [6]. Furthermore, inspired by the observation that quantization error often exhibits low-rank characteristics [13], we formulate the compensated output $\hat{y}_q$ as:

$$\hat{y}_q = y_q + UVx + b, \quad (9)$$

where x is the submodule input, and $U \in \mathbb{R}^{d \times r}$, $V \in \mathbb{R}^{r \times d}$ constitute a low-rank adapter with rank $r \ll d$. The optimal compensation matrix $W_{comp} \approx UV$ and bias b is estimated via closed-form linear regression on the calibration set:

$$(W_{comp}, b) = \arg \min_{W, b} \|y - (y_q + Wx + b)\|_2^2. \quad (10)$$

Table 1: Quantization results on the ETH3D dataset. Best results in each column are highlighted in bold. Our method is highlighted in light yellow.

Model	Bit-Width	Method	Acc. ↓		Comp. ↓		N.C. ↑	
			Mean	Med.	Mean	Med.	Mean	Med.
VGGT	W4A8	RTN	0.879	0.868	3.504	2.827	0.543	0.568
		PTQ4ViT	0.669	0.559	1.107	0.770	0.684	0.765
		ERQ	1.010	0.954	2.368	1.978	0.547	0.582
		RepQ	1.144	1.112	2.895	2.311	0.535	0.548
		GPTQ	0.872	0.834	2.686	2.046	0.539	0.563
		Ours	0.492	0.367	0.677	0.446	0.761	0.876
VGGT	W6A6	RTN	0.900	0.804	1.965	1.350	0.572	0.611
		PTQ4ViT	0.389	0.254	0.440	0.276	0.806	0.914
		ERQ	0.598	0.455	0.679	0.358	0.737	0.839
		RepQ	0.946	0.855	2.222	1.547	0.577	0.612
		GPTQ	0.407	0.290	0.549	0.345	0.783	0.889
		Ours	0.334	0.216	0.406	0.226	0.813	0.925
Pi3	W4A8	RTN	1.181	1.178	2.281	1.669	0.522	0.534
		PTQ4ViT	0.413	0.289	0.480	0.307	0.780	0.898
		RepQ	0.976	0.940	3.062	2.514	0.513	0.514
		ERQ	0.869	0.833	2.802	2.173	0.559	0.585
		GPTQ	0.872	0.834	2.686	2.046	0.539	0.563
		Ours	0.392	0.271	0.478	0.308	0.781	0.907
Pi3	W6A6	RTN	1.134	1.135	2.829	2.095	0.525	0.534
		PTQ4ViT	0.221	0.140	0.237	0.136	0.845	0.955
		RepQ	0.888	0.844	2.398	1.853	0.532	0.548
		ERQ	0.293	0.202	0.270	0.141	0.743	0.858
		GPTQ	0.407	0.290	0.549	0.345	0.783	0.889
		Ours	0.209	0.130	0.228	0.131	0.856	0.960

To minimize deployment overhead, we apply truncated Singular Value Decomposition (SVD) to W_{comp} and retain only the r dominant singular components to construct U and V . This sequential estimation ensures that each module’s compensation accounts for errors propagated from all preceding layers.

Tail Relative Error (TRE) Gating. While module-wise compensation rectifies local bias, indiscriminate application to all layers incurs unnecessary parameter overhead and risks over-correction. Crucially, as shown in Fig. 5, quantization distortions are not uniformly distributed but are often dominated by high-magnitude activation outliers. Conventional global metrics like MSE fail to capture this sensitivity as they dilute the impact of outliers across the entire feature dimension.

We therefore introduce **Tail Relative Error (TRE)** to identify modules where high-magnitude distortions significantly compromise geometric fidelity. For a module output $y \in \mathbb{R}^n$, we define the tail index set $\mathcal{S}$ containing the indices of the top- k elements with the largest absolute magnitudes, where $k = \lfloor \rho n \rfloor$ (typically $\rho = 0.01$). The TRE is defined as:

$$\text{TRE}(y, y_q) = \frac{\sum_{i \in \mathcal{S}} (y_i - y_{q,i})^2}{\sum_{i \in \mathcal{S}} y_i^2 + \epsilon}, \quad (11)$$

where ϵ is a small constant for numerical stability. TRE explicitly quantifies the relative energy distortion concentrated in the long-tailed regions of the activation distribution. We employ TRE as an adaptive gating mechanism: the compensation branch (U, V) is activated only if $\text{TRE} > \tau$, where τ is a predefined threshold. This selective activation ensures that hardware resources are prioritized for the most error-sensitive modules, preserving geometric consistency with minimal computational footprint.

Table 2: Quantization results on the 7Scenes dataset. Best results in each column are highlighted in bold. Our method is highlighted in light yellow .

Model	Bit-Width	Method	Acc. ↓		Comp. ↓		N.C. ↑	
			Mean	Med.	Mean	Med.	Mean	Med.
VGGT	W4A8	RTN	0.059	0.038	0.657	0.575	0.586	0.631
		PTQ4ViT	0.046	0.022	0.056	0.022	0.680	0.776
		ERQ	0.077	0.044	0.371	0.273	0.610	0.664
		RepQ	0.078	0.051	0.425	0.334	0.611	0.666
		GPTQ	0.086	0.057	0.422	0.306	0.594	0.646
		Ours	0.028	0.011	0.044	0.015	0.687	0.788
	W6A6	RTN	0.093	0.055	0.165	0.070	0.637	0.708
		PTQ4ViT	0.023	0.009	0.039	0.015	0.690	0.792
		ERQ	0.048	0.024	0.067	0.029	0.671	0.764
		RepQ	0.127	0.082	0.180	0.075	0.594	0.645
		GPTQ	0.022	0.009	0.038	0.014	0.689	0.792
		Ours	0.021	0.009	0.037	0.014	0.690	0.793
Pi3	W4A8	RTN	0.043	0.033	0.712	0.643	0.497	0.496
		PTQ4ViT	0.027	0.013	0.044	0.014	0.674	0.771
		RepQ	0.069	0.047	0.451	0.344	0.493	0.490
		ERQ	0.068	0.045	0.426	0.326	0.536	0.554
		GPTQ	0.086	0.057	0.422	0.306	0.594	0.646
		Ours	0.026	0.013	0.046	0.016	0.672	0.768
	W6A6	RTN	0.057	0.043	0.568	0.481	0.503	0.504
		PTQ4ViT	0.018	0.008	0.022	0.009	0.683	0.785
		RepQ	0.096	0.060	0.213	0.087	0.508	0.511
		ERQ	0.021	0.010	0.017	0.006	0.564	0.598
		GPTQ	0.022	0.009	0.038	0.014	0.689	0.792
		Ours	0.017	0.007	0.022	0.009	0.681	0.782

4 Experiments

4.1 Experiment and Evaluation Settings

Base Model. We evaluate VGGT-1B [24] and Pi3 [26], and quantize each model directly from its pre-trained full-precision checkpoint without retraining.

Tasks and Datasets. We mainly evaluate point cloud reconstruction. For calibration, we randomly sample 20 instances from the DTU training split and construct an 8-sample calibration set using our Progressive Calibration Set Construction; this set is used for all experiments. For evaluation, we report reconstruction results on 7Scenes and ETH3D. We also evaluate the prediction of camera parameters on Co3Dv2 to test cross-task robustness.

Quantization Settings. We consider three common low-bit configurations: W4A8, W6A6, and W8A8.

Baseline Methods. We compare with representative PTQ baselines, including Round-to-nearest (RTN), PTQ4ViT [32], RepQ-ViT [14], ERQ, and GPTQ [5]. For GPTQ, whose original implementation quantizes weights only, we adopt PT4ViT activation quantizer.

For a fair and controlled comparison, we evaluate all methods under a unified *per-tensor* weight quantization setting. Our method is inherently agnostic to the choice of quantization granularity and can be applied under both per-tensor and channel-wise settings, whereas several baseline methods are originally formulated with channel-wise weights. Allowing channel-wise quantization for these baselines in the main comparison would introduce an inherent advantage due to the additional per-channel scaling flexibility. We therefore standardize all baselines to the per-tensor setting, ensuring that the comparison primarily reflects differences

Table 3: Camera parameter prediction results on VGGT. Best results in each column are highlighted in bold. Our method is highlighted in light yellow .

Bit-Width	Method	A@30↑	A@15↑	A@5↑	A@3↑
W4A8	RTN	0.0041	0.0007	0.0000	0.0000
	PTQ4ViT	0.7622	0.6151	0.2753	0.1363
	GPTQ	0.0334	0.0137	0.0010	0.0005
	ERQ	0.0202	0.0043	0.0006	0.0002
	REPQ	0.0063	0.0006	0.0000	0.0000
	Ours	0.8099	0.6848	0.3975	0.2484
W8A8	RTN	0.8998	0.8190	0.5914	0.4380
	PTQ4ViT	0.9220	0.8592	0.6877	0.5635
	GPTQ	0.9225	0.8608	0.6907	0.5686
	ERQ	0.9014	0.8193	0.5596	0.3906
	REPQ	0.8524	0.7373	0.4379	0.2827
	Ours	0.9252	0.8620	0.6933	0.5723

in calibration and error-mitigation strategies rather than quantizer granularity. For completeness, we additionally compare our method with channel-wise variants of these baselines in Appendix 4.

Implementation Details. Unless otherwise specified, all methods use the same calibration data and inference settings. We use a tail ratio of $\rho=0.01$ for **Tail Relative Error (TRE)**, set the compensation threshold to $\tau=0.007$, and fix the rank of the low-rank compensation module to $r=16$.

Metrics. For point cloud reconstruction, we report Acc. and Comp. (lower is better) and N.C. (higher is better). For camera parameter prediction on Co3Dv2, we report AUC at multiple thresholds (A@30/A@15/A@5/A@3; higher is better).

4.2 Main Results

Quantitative Comparison. We evaluate TAPTQ on point cloud reconstruction across 7Scenes and ETH3D under two quantization settings, W4A8 and W6A6. Results are summarized in Tables 2 and 1. Lower Acc. and Comp. indicate better reconstruction quality, while higher N.C. reflects improved normal consistency.

Overall, TAPTQ consistently improves reconstruction performance across benchmarks and architectures, with larger gains observed under aggressive quantization and on more challenging scenes. On 7Scenes, TAPTQ achieves the best or competitive results on both VGGT and Pi3. Notably, under W4A8, TAPTQ clearly outperforms existing PTQ baselines on VGGT, while under W6A6, where performance becomes saturated, it remains among the top-performing methods without degradation. On ETH3D, TAPTQ delivers more pronounced improvements across both models, highlighting its robustness to complex geometry and severe quantization noise.

Qualitative Comparison. Figure 6 compares the W6A6 reconstruction results produced by PTQ4ViT and TAPTQ on VGGT and Pi3. Under the same quantization setting, TAPTQ better preserves geometric structures and surface continuity.

Camera Parameter Prediction. Table 3 reports camera parameter prediction results on VGGT under W4A8 and W8A8. This task is highly sensitive to quantization. Under W4A8, most baselines collapse, while TAPTQ substantially improves AUC across thresholds. Under W8A8, TAPTQ achieves the best performance.

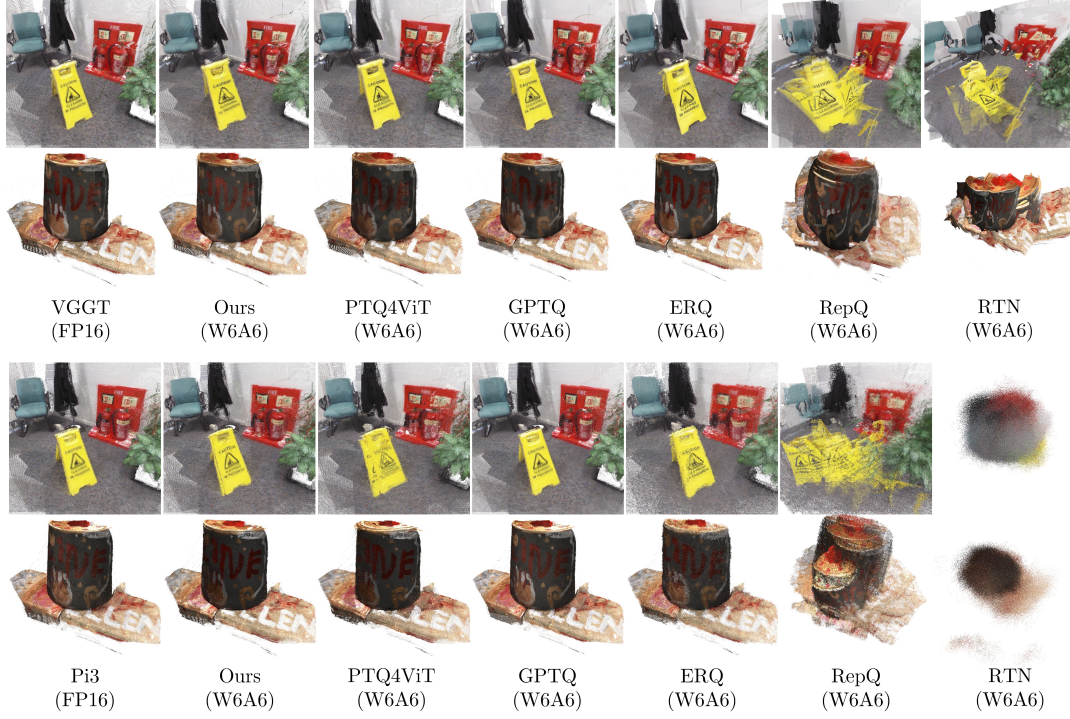


Figure 6: Qualitative Comparisons. Results on VGGT (top) and Pi3 (bottom).

Notably, although calibration is conducted on DTU, TAPTQ still transfers effectively to this additional camera parameter prediction task, demonstrating strong generalization beyond the calibration data and target metric.

4.3 Ablation Study

Effect of Calibration Set Construction. Table 5 compares calibration set construction strategies on 7Scenes under W4A8, using the same initial pool of 20 DTU training samples. *All samples* uses all 20 samples as a high-cost reference, while *rand* reports multiple random draws to reflect selection variance. *Stability score* follows the noise-filtering principle of QuantVGGT, filtering unstable samples based on feature statistics and then performing $K=4$ k -means with proportional sampling.

As shown in Table 5, random selection exhibits large variance under limited budgets. With only 8 samples, our method achieves the best overall performance and even surpasses the 20-sample baseline, highlighting the importance of filtering out noisy samples.

Effect of the Compensation Threshold τ . We further investigate the impact of the compensation threshold τ in the proposed TRE-Guided Compensation Module. Table 6 presents the results on 7Scenes under W4A8. As shown in Table 6, quantization without compensation causes a substantial degradation in reconstruction quality. Full compensation recovers performance but may over-correct quantization-insensitive modules. In contrast, TRE-guided compensation provides better performance by selectively compensating only high-TRE modules. As τ increases from 0.005 to 0.02,

Table 4: Channel-wise quantization results. Results on the 7Scenes dataset using the VGGT model. Best results in each column are highlighted in bold. Our method is highlighted in light yellow .

Bit-Width	Method	Acc.↓		Comp.↓		N.C.↑	
		Mean	Med.	Mean	Med.	Mean	Med.
W4A8	ERQ (ch-wise)	0.026	0.012	0.043	0.018	0.683	0.783
	RepQ (ch-wise)	0.058	0.027	0.100	0.032	0.661	0.747
	GPTQ (ch-wise)	0.021	0.009	0.033	0.017	0.686	0.788
	Ours (ch-wise)	0.018	0.008	0.032	0.015	0.688	0.790
W6A6	ERQ (ch-wise)	0.033	0.014	0.045	0.018	0.675	0.773
	RepQ (ch-wise)	0.037	0.016	0.048	0.019	0.680	0.778
	GPTQ (ch-wise)	0.021	0.009	0.036	0.015	0.691	0.794
	Ours (ch-wise)	0.020	0.008	0.036	0.015	0.692	0.796

the number of compensated modules decreases; $\tau=0.007$ achieves the best overall performance.

In terms of storage overhead, the proposed compensation module introduces only a marginal increase in model size. Specifically, the size grows from 608.08 MB (no compensation) to around 610–616 MB with compensation, corresponding to an overhead of less than 2%. Compared to the FP32 model (approximately 4 GB), the additional cost remains negligible while achieving substantial compression.

Effect of Ternary Search and Compensation. Table 7 evaluates ternary search and module-wise compensation under W6A6. Ternary search replaces exhaustive interval enumeration, substantially reducing calibration time with comparable accuracy. Module-wise

Table 5: Ablation on calibration set construction. Model: VGGT, dataset: 7Scenes, setting: W4A8. All methods select calibration samples from the same initial pool of 20 DTU training samples. Best results in each column are highlighted in bold. Our method is highlighted in light yellow .

Size	Method	Acc.↓		Comp.↓		N.C.↑	
		Mean	Med.	Mean	Med.	Mean	Med.
20	All samples	0.0281	0.0123	0.0452	0.0161	0.6864	0.7873
	rand (case1)	0.0370	0.0146	0.0580	0.0241	0.6842	0.7835
	rand (case2)	0.0354	0.0158	0.0495	0.0198	0.6864	0.7863
	rand (case3)	0.0332	0.0134	0.0508	0.0178	0.6851	0.7856
	rand (case4)	0.0307	0.0122	0.0468	0.0165	0.6836	0.7819
	Stability Score	0.0291	0.0122	0.0470	0.0161	0.6851	0.7852
4	Ours	0.0295	0.0121	0.0460	0.0166	0.6868	0.7875
	rand (case1)	0.0300	0.0131	0.0496	0.0177	0.6868	0.7884
	rand (case2)	0.0342	0.0139	0.0449	0.0158	0.6818	0.7806
	rand (case3)	0.0318	0.0134	0.0478	0.0169	0.6842	0.7841
	Stability Score	0.0330	0.0137	0.0436	0.0164	0.6811	0.7805
	Ours	0.0264	0.0105	0.0444	0.0147	0.6867	0.7884

compensation further improves reconstruction quality with marginal overhead, indicating that the two components are complementary. Specifically, enabling ternary search reduces the calibration time from 522.6 to 161.5 minutes (3.2× speedup) with negligible accuracy change (Acc. 0.023 vs. 0.023). Adding module-wise compensation on top further lowers Acc. from 0.023 to 0.021 and improves N.C. from 0.688 to 0.692, at a cost of only 7.2 additional minutes, confirming that the two components jointly achieve a favorable efficiency–accuracy trade-off.

4.4 Discussion

Performance Gap Analysis. A consistent trend across our experiments is that TAPTQ’s advantage becomes more pronounced under more challenging settings. Under aggressive quantization (W4A8), TAPTQ achieves substantially larger gains over baselines than under moderate settings (W6A6). For instance, on ETH3D with VGGT at W4A8, TAPTQ reduces mean Acc. by 26% relative to the strongest baseline, whereas the gap narrows considerably at W6A6. Similarly, on the camera parameter prediction task at W4A8, most baselines nearly collapse (e.g., RTN achieves only 0.004 A@30), while TAPTQ maintains 0.81. This suggests that the tail-aware error correction mechanism specifically targets the most vulnerable aspect of 3D model quantization—the amplification of distortions in high-magnitude activation regions—and its benefit scales with quantization difficulty.

Cross-task Generalization. It is worth noting that TAPTQ’s calibration set is constructed entirely from DTU, yet the quantized models generalize well to both point cloud reconstruction (7Scenes, ETH3D) and camera parameter prediction (Co3Dv2). This cross-task transferability suggests that the progressive calibration strategy captures geometrically representative samples rather than overfitting to a specific data distribution or evaluation metric, which is practically appealing as it eliminates the need for task-specific recalibration during deployment.

Table 6: Ablation on compensation threshold τ under W4A8. Results on 7Scenes.

Strategy	Acc.↓		Comp.↓		N.C.↑		Size (MB)
	Mean	Med.	Mean	Med.	Mean	Med.	
No comp	0.0457	0.0218	0.0562	0.0222	0.6795	0.7757	608.0799
Full comp	0.0291	0.0117	0.0377	0.0147	0.6863	0.7876	616.9549
$\tau=0.005$	0.0271	0.0110	0.0392	0.0146	0.6835	0.7842	612.2049
$\tau=0.01$	0.0279	0.0110	0.0436	0.0153	0.6868	0.7883	610.3299
$\tau=0.02$	0.0306	0.0125	0.0489	0.0157	0.6872	0.7883	609.3924
$\tau=0.007$	0.0251	0.0102	0.0401	0.0146	0.6878	0.7903	611.3299

Table 7: Effect of ternary search and module-wise compensation under W6A6. Quantitative results on the 7Scenes dataset that evaluate the reconstruction performance and calibration efficiency. All results report mean values; Time denotes calibration time.

Ternary Search	Error Compens.	Acc.↓	Comp.↓	N.C.↑	Time
✓	✓	0.021	0.035	0.692	168.7 min
✓	✗	0.023	0.040	0.688	161.5 min
✗	✓	0.020	0.036	0.691	529.7 min
✗	✗	0.023	0.039	0.690	522.6 min

Limitations. Despite its effectiveness, TAPTQ has several limitations worth noting. First, it has been validated on only two 3D geometry foundation models (VGGT and Pi3); its generalizability to architecturally diverse 3D models remains to be verified. Second, although the near-unimodal property of the similarity landscape is empirically supported (Fig. 3), a formal theoretical guarantee is lacking. Under extreme quantization or in certain atypical layers, multi-modal behavior may arise, potentially causing the ternary search to converge to a local optimum. Third, as a PTQ method, the accuracy gap between TAPTQ and Quantization-Aware Training (QAT) has not been quantified. Integrating the tail-aware philosophy into QAT frameworks presents a promising direction for future work.

5 Conclusion

We propose **TAPTQ**, a Tail-Aware Post-Training Quantization pipeline tailored for 3D geometry models. TAPTQ combines **Progressive Calibration Set Construction** for stable range estimation under limited calibration budgets, **Quantization Interval Optimization via Ternary Search** to reduce interval calibration cost, and **TRE-Guided Module-wise Compensation** to selectively correct cross-layer error accumulation. Across point cloud reconstruction benchmarks (7Scenes/ETH3D) and camera parameter prediction on Co3Dv2, TAPTQ consistently improves quantized performance under low-bit settings (e.g., W4A8/W6A6/W8A8), with particularly strong gains under aggressive quantization. Ablations further validate the complementarity of ternary search and TRE-guided compensation, showing improved calibration efficiency and accuracy–consistency trade-offs.

References

- [1] Ron Banner, Yury Nahshan, and Daniel Soudry. 2019. Post training 4-bit quantization of convolutional networks for rapid-deployment. *Advances in neural information processing systems* 32 (2019).
- [2] Jonathan T Barron, Ben Mildenhall, Matthew Tancik, Peter Hedman, Ricardo Martin-Brualla, and Pratul P Srinivasan. 2021. Mip-nerf: A multiscale representation for anti-aliasing neural radiance fields. In *Proceedings of the IEEE/CVF international conference on computer vision*. 5855–5864.
- [3] Hainan Cui, Xiang Gao, Shuhan Shen, and Zhanyi Hu. 2017. HSfM: Hybrid structure-from-motion. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 1212–1221.
- [4] Weilun Feng, Haotong Qin, Mingqiang Wu, Chuanguang Yang, Yuqi Li, Xiangqi Li, Zhulin An, Libo Huang, Yulun Zhang, Michele Magno, et al. 2025. Quantized Visual Geometry Grounded Transformer. *arXiv preprint arXiv:2509.21302* (2025).
- [5] Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. 2022. Gptq: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323* (2022).
- [6] Minghao Fu, Hao Yu, Jie Shao, Junjie Zhou, Ke Zhu, and Jianxin Wu. 2025. Quantization without tears. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 4462–4472.
- [7] Amir Gholami, Sehoon Kim, Zhen Dong, Zhewei Yao, Michael W Mahoney, and Kurt Keutzer. 2022. A survey of quantization methods for efficient neural network inference. In *Low-power computer vision*. Chapman and Hall/CRC, 291–326.
- [8] Bowei He, Lihao Yin, Huiling Zhen, Shuqi Liu, Han Wu, Xiaokun Zhang, Mingxuan Yuan, and Chen Ma. 2025. Preserving LLM Capabilities through Calibration Data Curation: From Analysis to Optimization. *arXiv preprint arXiv:2510.10618* (2025).
- [9] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* 42, 4 (2023), 139–1.
- [10] Vincent Leroy, Yohann Cabon, and Jérôme Revaud. 2024. Grounding image matching in 3d with mast3r. In *European Conference on Computer Vision*. Springer, 71–91.
- [11] Xiuyu Li, Yijiang Liu, Long Lian, Huanrui Yang, Zhen Dong, Daniel Kang, Shanghang Zhang, and Kurt Keutzer. 2023. Q-diffusion: Quantizing diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 17535–17545.
- [12] Yuhang Li, Ruihao Gong, Xu Tan, Yang Yang, Peng Hu, Qi Zhang, Fengwei Yu, Wei Wang, and Shi Gu. 2021. Brecq: Pushing the limit of post-training quantization by block reconstruction. *arXiv preprint arXiv:2102.05426* (2021).
- [13] Yixiao Li, Yifan Yu, Chen Liang, Pengcheng He, Nikos Karampatziakis, Weizhu Chen, and Tuo Zhao. 2023. Loftq: Lora-fine-tuning-aware quantization for large language models. *arXiv preprint arXiv:2310.08659* (2023).
- [14] Zhikai Li, Junrui Xiao, Lianwei Yang, and Qingyi Gu. 2023. Repq-vit: Scale reparameterization for post-training quantization of vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 17227–17236.
- [15] Haotong Lin, Sili Chen, Junhao Liew, Donny Y Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. 2025. Depth anything 3: Recovering the visual space from any views. *arXiv preprint arXiv:2511.10647* (2025).
- [16] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. 2024. Awq: Activation-aware weight quantization for on-device llm compression and acceleration. *Proceedings of machine learning and systems* 6 (2024), 87–100.
- [17] Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. 2021. Nerf: Representing scenes as neural radiance fields for view synthesis. *Commun. ACM* 65, 1 (2021), 99–106.
- [18] Markus Nagel, Rana Ali Amjad, Mart Van Baalen, Christos Louizos, and Tijmen Blankevoort. 2020. Up or down? adaptive rounding for post-training quantization. In *International conference on machine learning*. PMLR, 7197–7206.
- [19] Markus Nagel, Mart van Baalen, Tijmen Blankevoort, and Max Welling. 2019. Data-free quantization through weight equalization and bias correction. In *Proceedings of the IEEE/CVF international conference on computer vision*. 1325–1334.
- [20] Johannes L Schonberger and Jan-Michael Frahm. 2016. Structure-from-motion revisited. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 4104–4113.
- [21] You Shen, Zhipeng Zhang, Yansong Qu, Xiawu Zheng, Jiayi Ji, Shengchuan Zhang, and Liujuan Cao. 2025. Fastvggt: Training-free acceleration of visual geometry transformer. *arXiv preprint arXiv:2509.02560* (2025).
- [22] Chen Tang, Kai Ouyang, Zhi Wang, Yifei Zhu, Wen Ji, Yaowei Wang, and Wenwu Zhu. 2022. Mixed-precision neural network quantization via learned layer-wise importance. In *European conference on computer vision*. Springer, 259–275.
- [23] Chung-Shien Brian Wang, Christian Schmidt, Jens Piekenbrinck, and Bastian Leibe. 2025. Faster vgg with block-sparse global attention. *arXiv preprint arXiv:2509.07120* (2025).
- [24] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. 2025. Vggt: Visual geometry grounded transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 5294–5306.
- [25] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. 2024. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 20697–20709.
- [26] Yifan Wang, Jianjun Zhou, Haoyi Zhu, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, and Tong He. 2025. π^3 : Permutation-Equivariant Visual Geometry Learning. *arXiv preprint arXiv:2507.13347* (2025).
- [27] Xiying Wei, Ruihao Gong, Yuhang Li, Xianglong Liu, and Fengwei Yu. 2022. Qdrop: Randomly dropping quantization for extremely low-bit post-training quantization. *arXiv preprint arXiv:2203.05740* (2022).
- [28] Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. 2023. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International conference on machine learning*. PMLR, 38087–38099.
- [29] Shuzhao Xie, Jiahang Liu, Weixiang Zhang, Shijia Ge, Sicheng Pan, Chen Tang, Yunpeng Bai, Cong Zhang, Xiaoyi Fan, and Zhi Wang. 2025. SizeGS: Size-aware Compression of 3D Gaussian Splatting via Mixed Integer Programming. In *ACM MM*.
- [30] Shuzhao Xie, Weixiang Zhang, Chen Tang, Yunpeng Bai, Rongwei Lu, Shijia Ge, and Zhi Wang. 2024. MesonGS: Post-training Compression of 3D Gaussians via Efficient Attribute Transformation. In *European Conference on Computer Vision*. Springer.
- [31] Yao Yao, Zixin Luo, Shiwei Li, Tian Fang, and Long Quan. 2018. Mvsnet: Depth inference for unstructured multi-view stereo. In *Proceedings of the European conference on computer vision (ECCV)*. 767–783.
- [32] Zhihang Yuan, Chenhao Xue, Yiqi Chen, Qiang Wu, and Guangyu Sun. 2022. Ptg4vit: Post-training quantization for vision transformers with twin uniform quantization. In *European conference on computer vision*. Springer, 191–207.